



Fig. 5: An 8-channel relay module

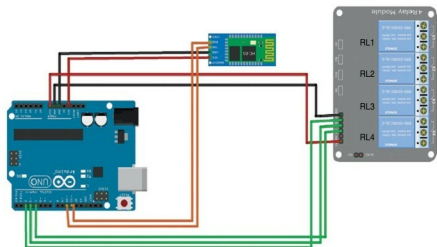


Fig. 6: Relay module connection

Arduino Uno is based on AT-mega328 microcontroller (MCU). It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board MCU, a power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the MCU. It is very easy to use as you simply need to connect it to a computer using a USB cable, or power it with an AC-to-DC adaptor or battery to get started. The MCU onboard is programmed in Arduino programming language using Arduino IDE.

Pins 10 and 11 of Arduino are connected to pins  $T_{XD}$  and  $R_{XD}$  of the Bluetooth module, respectively, as shown in Fig. 6. Pins Gnd and Vcc of the Bluetooth module are connected to Gnd and +3.3V of Arduino board respectively. Pins 2, 3 and 4 are connected to the three relays (RL1, RL2 and RL3) of the relay board. Pins Vin and Gnd of the relay board are connected to pins Vin and Gnd of Arduino board, respectively.

**Note.** Vin is usually used to give input power, but since we are supplying 12V to Arduino using an adaptor, we can use Vin pin on Arduino to power the 12V relay module.

## Software

The software program (homeautomation.ino) is written in Arduino programming language called Processing. Arduino Uno is programmed using Arduino IDE software that you can download from [arduino.cc](http://arduino.cc). MIT App Inventor software was used to create the Android app (.apk) for this project.

The app on your smartphone sends data when you click on buttons or feed voice commands via Bluetooth in the mobile to Bluetooth module HC-05 connected with Arduino board.

Received data pin  $T_{XD}$  of the HC-05 is connected to Arduino. Arduino Uno processes the received data and controls the relay board accordingly.

Procedure for installing the Android app (.apk) is as follows:

1. Download the app (homeautomation.apk) from this month's EFY DVD or website. Copy it to your Android smartphone.
2. Run .apk file. It will prompt you to complete the action. Click Package Installer and then Install.
3. You will also need a voice-recognition app on your Android smartphone. Most smartphones have this app preinstalled. If you do not have it, download one from Google Play Store.

## Construction and testing

Assemble the circuit as shown in the circuit diagram. Open Arduino IDE and compile the program (sketch). Upload the sketch (homeautomation.ino) to Arduino board. Switch on the power supply to Arduino by connecting it to 12V power source. Pair Bluetooth module with your Android phone. Type password '1234' (default password) of Bluetooth module.

Click Bluetooth Image on the app to connect it with the Bluetooth module. It automatically connects and displays as Connected in the app.

You are now ready to control the appliances using the app. You can either use on/off buttons or voice commands to control the appliances. You can control more electrical appliances by increasing the number of channels in the relay. For instance, using an 8-channel relay, you can control up to eight devices. For this, you need to alter the source code by adding input commands and voice commands to control the devices. **TRY**

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)



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